Dear Members of the Selection Committee,

I am writing to express my strong motivation to pursue the **Erasmus Mundus Joint Master's in Data Engineering and Artificial Intelligence (DEAI)**. My academic training in computer science through a **Bachelor's degree in Information Management** from **Tribhuvan University**, together with professional experience as a Python developer, has prepared me well for advanced interdisciplinary study. Courses such as **Artificial Intelligence, Data Structures and Algorithms, Discrete Mathematics, Digital Logic Systems, Computer Organization, and Advanced Internetworking** provided a solid theoretical foundation, while my independent completion of **Harvard's CS50 Artificial Intelligence course** strengthened my understanding of machine learning concepts.

The relevance of DEAI is particularly significant in the context of **Nepal**, where digital transformation is growing but remains constrained by limited data infrastructure and lack of advanced analytical systems. Key sectors such as **healthcare, agriculture, and public governance** face challenges including inefficient resource allocation, absence of predictive analytics, and fragmented data management. These challenges highlight the urgent need for professionals skilled in **data engineering and artificial intelligence** who can design scalable data systems and intelligent models to support evidence-based decision-making. The DEAI program aligns directly with my aspiration to contribute to these solutions.

My **one year of professional experience as a Python developer**, working with **Python, SQL, and NumPy**, allowed me to engage with real-world data processing and problem solving. During my undergraduate AI coursework, I became particularly fascinated by machine learning's ability to learn from data and make intelligent decisions, motivating me to pursue deeper expertise in this field. My short-term goal is to develop strong expertise in machine learning for solving real-world problems, and my long-term goal is to work as an AI Engineer designing scalable and impactful AI-driven solutions.

The **DEAI program** offers an exceptional opportunity to develop advanced skills in data engineering and artificial intelligence within a research-driven international environment. Unlike programs that focus on AI or data engineering in isolation, DEAI uniquely integrates both domains, reflecting their strong interdependence in building scalable, real-world intelligent systems. I look forward to contributing to the **Erasmus Mundus community** by sharing my cultural perspective from Nepal and engaging actively in collaborative learning.

Thank you for considering my application.

Yours sincerely,

Sishir Gautam